9.1 Using Normal Distributions

Name _____ Period ____ Date _____

Algebra II 9.1 · Student sheet

I can…

- use the 68-95-99.7 rule to find areas under a normal curve.
- find probabilities for normally distributed data.
- tell whether data are approximately normal and how skew affects the mean and median.

Vocabulary · fill in the missing word

Normal distribution — a _____, symmetric distribution with mean μ and standard deviation σ

Normal curve — the bell-shaped graph of a normal _____; total area 1

68-95-99.7 rule — about 68%, 95%, and 99.7% of the data are within 1, 2, and 3 standard _____ of the mean

Skewed — not _____: the mean is pulled toward the tail

Part 1 · Coin-flip bell

Flip 10 coins. Number of heads: _____. Record the class results.

Heads	0	1	2	3	4	5	6	7	8	9	10
Students											

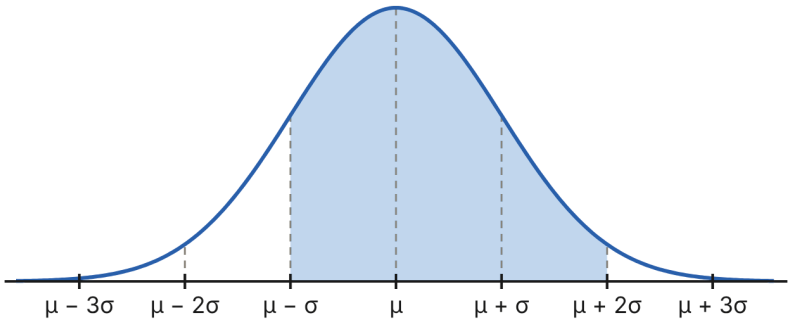
What shape is it? Where is the center? _____

Part 2 · Notes and examples (fill in each note, then work the example)

The 68-95-99.7 rule: Within 1σ of the mean: _____% within 2σ: _____% within 3σ: _____%

Example 1

What percent of the area under the normal curve is shaded?

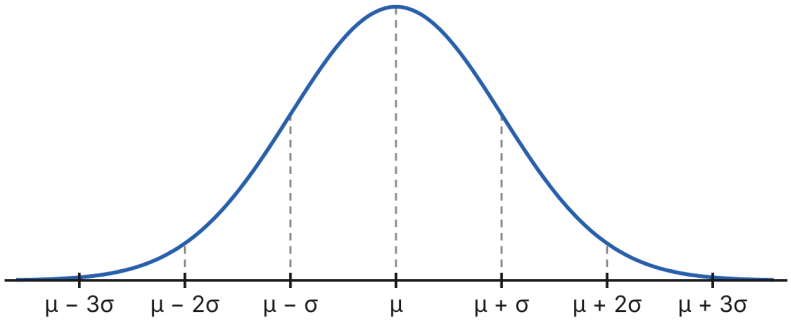


Areas of the bands: Bands on each side: μ to σ: _____%, σ to 2σ: _____%, 2σ to 3σ: _____%, beyond 3σ: _____%

Example 2

A normal distribution has mean μ and standard deviation σ. Find each probability.

a. $P(x \le \mu - \sigma)$ b. $P(x \ge \mu + 2\sigma)$



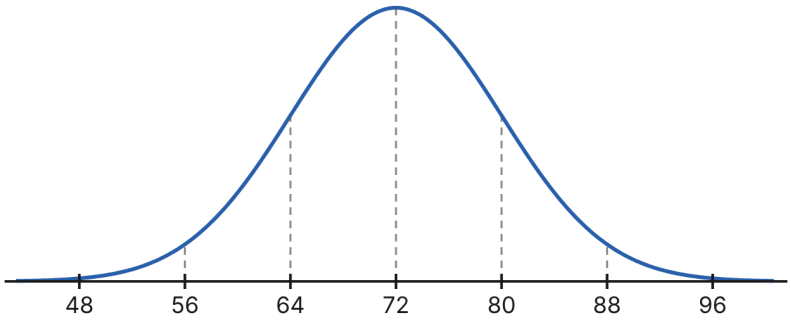
Example 3

Test scores are normally distributed with a mean of 72 and a standard deviation of 8.

a. What percent of the scores are between 64 and 80?

b. What percent are greater than 88?

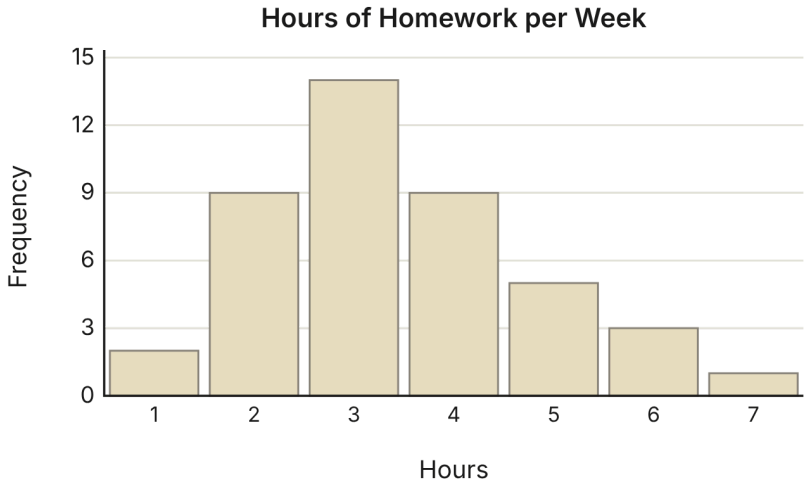
c. In a group of 500 students, about how many score greater than 88?



Skewed data: Skewed right: mean _____ median. Skewed left: mean _____ median.

Example 4

Does the histogram show data that are approximately normal? Is the mean greater or less than the median?



Part 3 · Partner practice

Partner 1

A normal distribution has a mean of 150 and a standard deviation of 10. Find P(x < 130).

Partner 2

What percent of the area under a normal curve is between $\mu - 3\sigma$ and $\mu - \sigma$?

